Classic 3 W / 1.2 W: Sprint calibrated: independent Fox 10 mW
NORMAL: protected 4S, 12.0–16.8 V. FOX–only: controller 3.3 V from 2S.
No 2 m changes. PRODUCTION–INTENT – NOT RELEASED. No measured prototype results.
Inter–sheet connections use explicit named global nets.
Read docs/M4A_REVIEW_DECISION.md and retained E2 safety gates.

01 Controller interface and local source

File: 01_source_interface.kicad_sch

02 Autonomous lease, inhibit and rearm

File: 02_interlocks.kicad_sch

03 Reduced permission logic and population–specific guards

File: 03_safety_logic.kicad_sch

04 Deliberate rearm, witnessed heartbeat and captured configuration

File: 04_arm_key.kicad_sch

05 4S input, one buck and independent rail isolation

File: 05_power.kicad_sch

06 Reference, pack voltage and thermal/current trips

File: 06_analog_protection.kicad_sch

06b Independent threshold dividers

File: 06b_thresholds.kicad_sch

07 PA current trip and case–temperature interlock

File: 07_current_thermal.kicad_sch

08 Proven normal RF core and QCX–derived drain shaping

File: 08_normal_rf.kicad_sch

08b Shared key–tail and disarm reset

File: 08b_key_tail.kicad_sch

09 Original 80 m elliptic filter and normal port

File: 09_normal_filter.kicad_sch

10 Independent Foxoring shaper and rail–following clock buffer

File: 10_fox_rf.kicad_sch

11 Separate Foxoring filter, calibration pad and protected compact port

File: 11_fox_filter.kicad_sch

12 Retained input damper and production test access

File: 12_evaluation.kicad_sch

PRODUCTION–INTENT – NOT RELEASED – NO HARDWARE RESULTS
ARDF–TX / review only

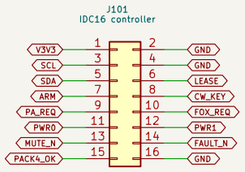
Sheet: /
File: RF80_V1.kicad_sch

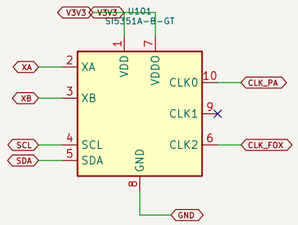
Title: RF80 V1 – production–intent, not released

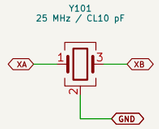
Size: A3	Date: 2026–10–18	**Rev: M4A–V1R1**
KiCad E.D.A. 10.0.6		Id: 1/15

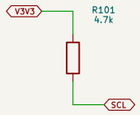
01 Controller interface and local source

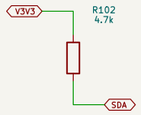
One local Si5351; no MCU/RTC/UI on RF PCB. Shared 3.3 V stays powered during inhibited I2C setup.
Controller wide-input regulator remains a proposal: evaluation fixture supplies bounded 3.3 V separately.

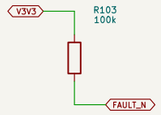


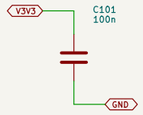


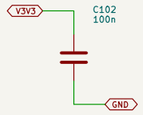


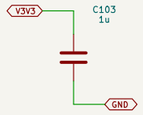


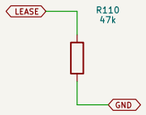


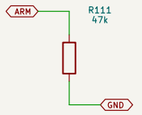


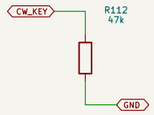


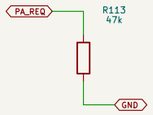


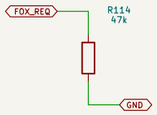


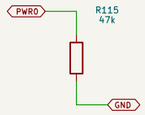


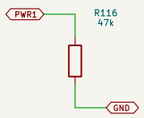


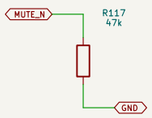


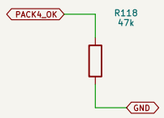












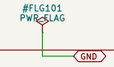












Logical contract: 3.3 V only; V3V3 3.135–3.465 V, absolute connector maximum 3.6 V.
RoutePAcurrentthroughdedicatedJ301return; checkcontrollerground–harnesscurrent. PA raw power has separate J301 and return.
LEASE 20 Hz software-progress toggles; ARM low before changing frequency, path or power.
Any SCL activity while armed trips the local latch. FAULT_N open drain, low when unarmed.
Do not connect TTL 5 V inputs or an unqualified controller raw supply to this connector.

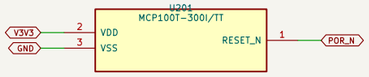
M4A-V1: PRODUCTION-INTENT – NOT RELEASED / all physical gates NOT_RUN.

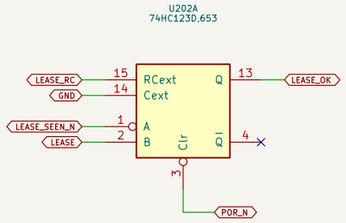
PRODUCTION-INTENT – NOT RELEASED – NO HARDWARE RESULTS
ARDF-TX
Sheet: /01 Controller interface and local source/
File: 01_source_interface.kicad_sch
Title: 01 Controller interface and local source

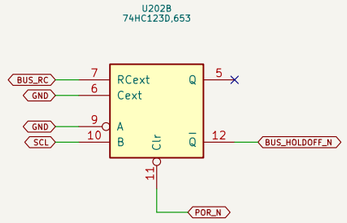
Size: A3	Date: 2026-10-18	**Rev: M4A-V1R1**
KiCad E.D.A. 10.0.6		Id: 2/15

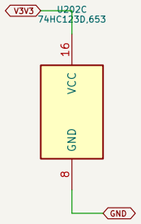
02 Autonomous lease, inhibit and rearm

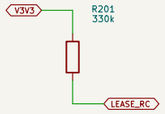
Healthy-only asynchronous reset. Latched fault cannot recover while ARM remains high.
ARM RC/Schmitt delay releases CLR before clocking. Requests exclusive; both asserted trips.

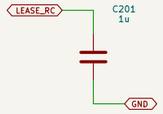


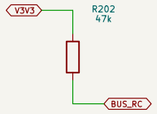


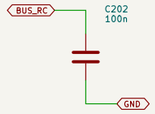


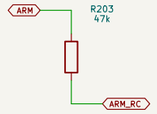


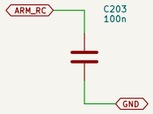


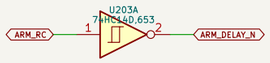












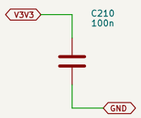


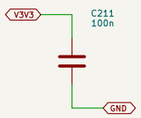


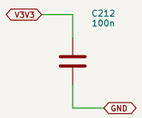


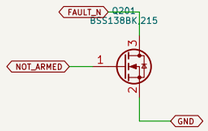


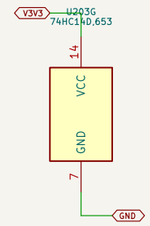












U202: first Q=pin13, /A=LEASE_SEEN_N; first validLEASE edge witnesses progress. Second /Q=pin12, /A=0,B=SCL.
No blind ISR heartbeat: controller advances LEASE only after schedule and bus-state checks.
Local mute is independent of firmware; hardware does not verify the requested frequency value.
Normal key-up leaves driver active through the measured drain-envelope tail. ARM/MUTE kill immediately.





M4A-V1: PRODUCTION-INTENT - NOT RELEASED / all physical gates NOT_RUN.

PRODUCTION-INTENT - NOT RELEASED - NO HARDWARE RESULTS
ARDF-TX

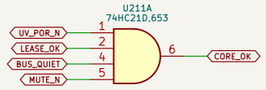
Sheet: /02 Autonomous lease, inhibit and rearm/
File: 02_interlocks.kicad_sch

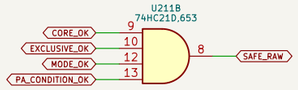
Title: 02 Autonomous lease, inhibit and rearm

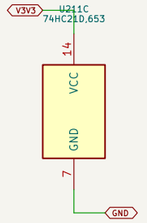
Size: A3	Date: 2026-10-18	**Rev: M4A-V1R1**
KiCad E.D.A. 10.0.6		Id: 3/15

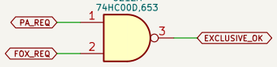
03 Reduced permission logic and population-specific guards

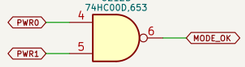
Four redundant gate stages are replaced by one core tree and wired open-collector PA trips.
Schottky logic margins and source loading remain E1 bench gates; no CMOS outputs are tied together.





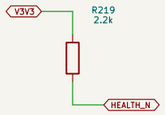


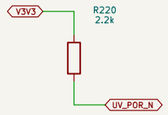


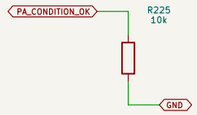


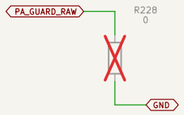


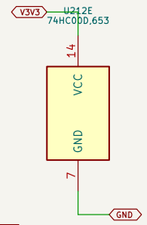














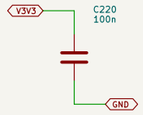


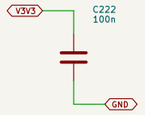












M4A-V1: PRODUCTION-INTENT - NOT RELEASED / all physical gates NOT_RUN.

PRODUCTION-INTENT - NOT RELEASED - NO HARDWARE RESULTS

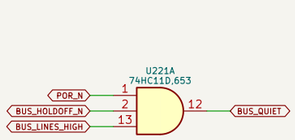
ARDF-TX

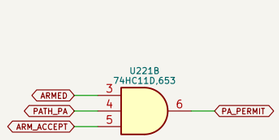
Sheet: /03 Reduced permission logic and population-specific guards/
File: 03_safety_logic.kicad_sch

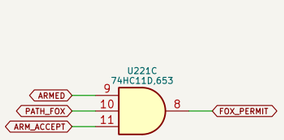
Title: 03 Reduced permission logic and population-specific guards

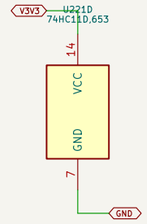
Size: A3	Date: 2026-10-18	**Rev: M4A-V1R1**
KiCad E.D.A. 10.0.6		Id: 4/15

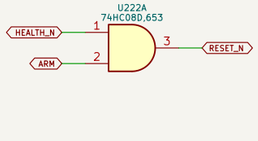
04 Deliberate rearm, witnessed heartbeat and captured configuration

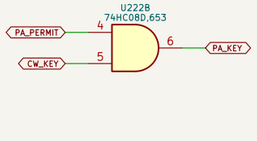


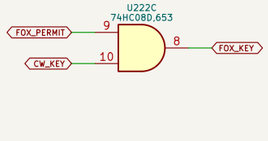


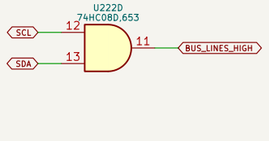


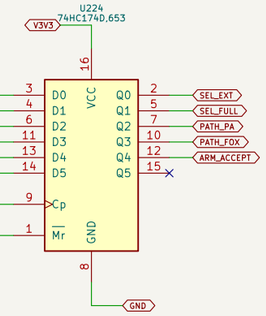


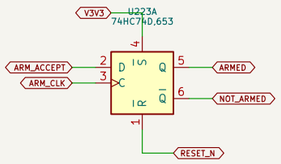


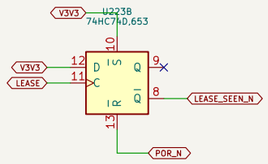


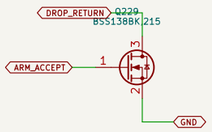


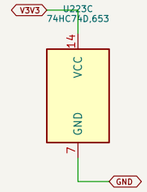


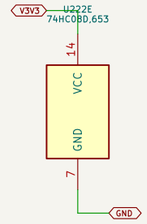


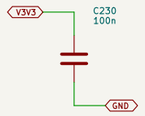


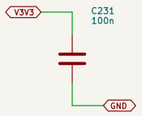


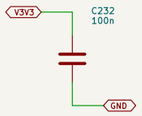


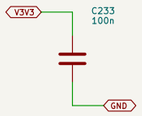




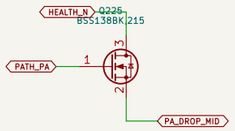


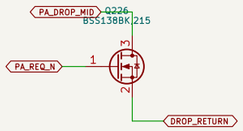


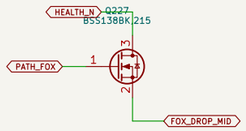


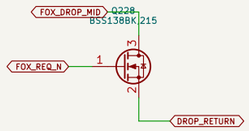


U224 captures path/mode/key-low on raw ARM; any health fault clears acceptance.
U223 delayed clock can only arm an accepted context. A fault in the pending delay cannot rearm.
Withdraw selected request: health clears context and RF permits. New raw ARM edge required.
Fresh ARM low>=20ms, configure requests/mode, then ARM high; wait>=250ms before first key.
Mode changes while armed are ignored; mode11, conflicting requests and SCL/SDA low fault immediately.
No recovery with ARM held high. U213/U214 deleted; their duplicated tree is not retained.









M4A-V1: PRODUCTION-INTENT - NOT RELEASED / all physical gates NOT_RUN.

PRODUCTION-INTENT - NOT RELEASED - NO HARDWARE RESULTS

ARDF-TX

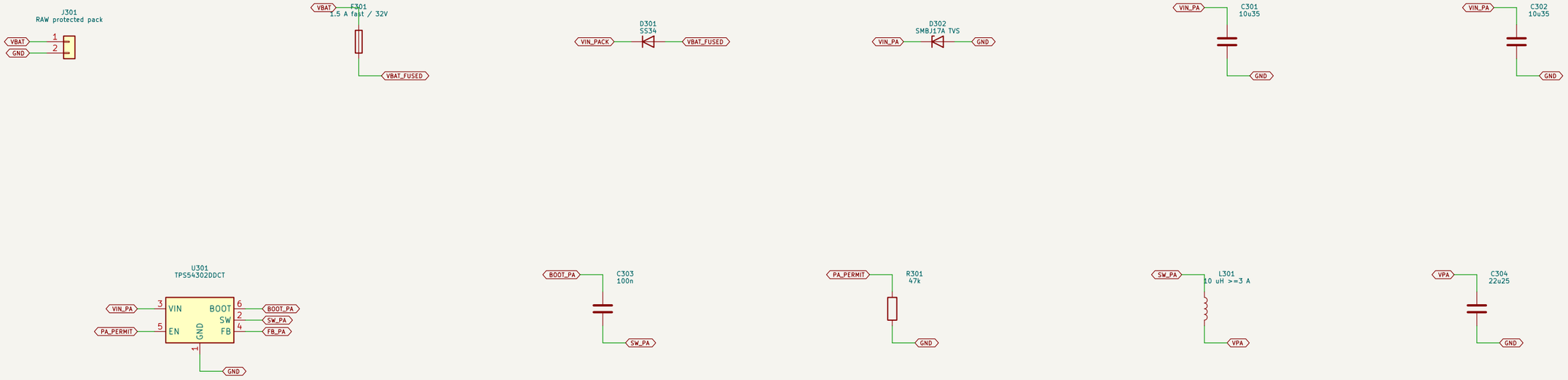
Sheet: /04 Deliberate rearm, witnessed heartbeat and captured configuration/
File: 04_arm_key.kicad_sch

Title: 04 Deliberate rearm, witnessed heartbeat and captured configuration

Size: A3	Date: 2026-10-18	**Rev: M4A-V1R1**
KiCad E.D.A. 10.0.6		Id: 5/15

05 4S input, one buck and independent rail isolation

Shared J301/F301/D301 input barrier; converter/loadswitch NORMAL only. Protected4S:12.0–16.8V.
Fox-only retains input barrier/sense bypass, omits buck/loadswitch and watt-rail parts;2S powerscontroller+input sense.
TPS22810 VIN must stay<=18V, including failed-buck transients;20V is absolute maximum.
28V passive simulation is NOT a system rating. Input clamp/fault coordination remains OPEN.



M4A-V1: PRODUCTION-INTENT – NOT RELEASED / all physical gates NOT_RUN.

PRODUCTION-INTENT – NOT RELEASED – NO HARDWARE RESULTS
ARDF-TX

Sheet: /05 4S input, one buck and independent rail isolation/
File: 05_power.kicad_sch

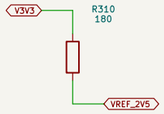
Title: 05 4S input, one buck and independent rail isolation

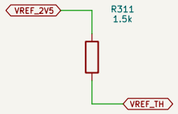
Size: A3	Date: 2026-10-18	**Rev: M4A-V1R1**
KiCad E.D.A. 10.0.6		Id: 6/15

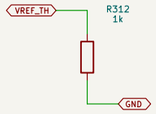
06 Reference, pack voltage and thermal/current trips

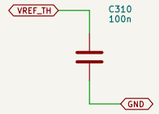
Real electrical trips; no RF power detector or continuous telemetry. Comparator inputs stay below1.5V.
Shared reverse-protected J301 pack input retained for Fox-only; all PA converter/driver/final components DNP.

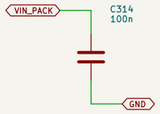


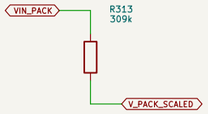


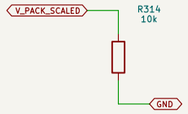


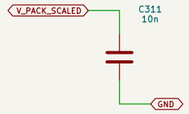


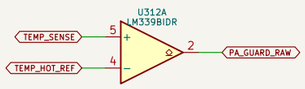


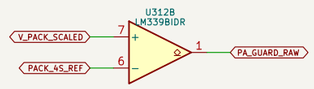


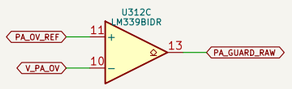


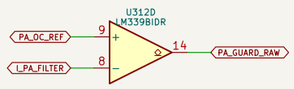


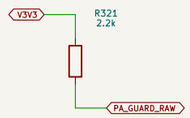


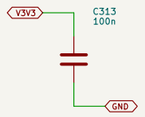


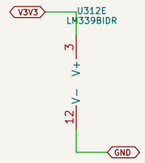


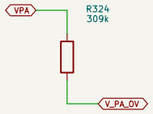


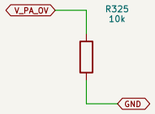












Nominal backup: commonUV5.810V/4Sbackup11.070V postreversebarrier: railOV8.973V.
Controller operational endpoint remains6.0/12.0V with hysteresis; cell-group BMS external.
Local pack sensing does not prove weakest-cell voltage. Hysteresis implemented by fault latch;
rearm only after input exceeds qualified margin. Fox-only: watt guards DNP, R228 clamps PA_GUARD_RAW; UV comparator remains local.

M4A-V1: PRODUCTION-INTENT - NOT RELEASED / all physical gates NOT_RUN.

PRODUCTION-INTENT - NOT RELEASED - NO HARDWARE RESULTS
ARDF-TX

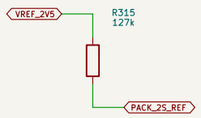
Sheet: /06 Reference, pack voltage and thermal/current trips/
File: 06_analog_protection.kicad_sch

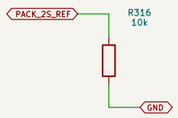
Title: 06 Reference, pack voltage and thermal/current trips

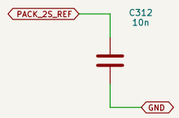
Size: A3	Date: 2026-10-18	**Rev: M4A-V1R1**
KiCad E.D.A. 10.0.6		Id: 7/15

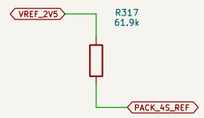
06b Independent threshold dividers

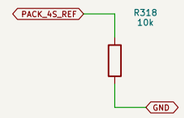
Independent reference dividers from regulated TL431 node, not a loaded shared half-reference.
Pack and PA voltage signals divideby31.9; INA current signal divides by4. No comparator pin exceeds1.05V
for28V input/rail or3.465V current-sense saturation. Thermal divider ratios sharetheir own<1.0V node.

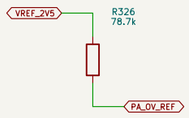


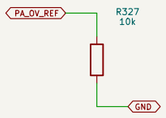


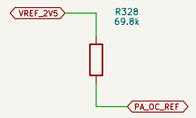


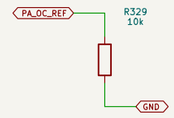












Calculated at2.495V reference: commonUV5.810V protectedinput;4Sbackup11.070V protectedinput;
OV8.973V rail; currenttrip1.251A. Addmeasured5S34drop forrawpackthreshold.
Controller operation stillstopsatapproximately6.0/12.0V; cell-group BMSexternal.
Referencebias180ohm supports>=1mA TL431cathodecurrentafterallparallelreferencebranchloads.
Cold/hotreference, comparatoroffset/inputbias, resistor tolerances andtripresponse mustbe measured.

M4A-V1: PRODUCTION-INTENT - NOT RELEASED / all physical gates NOT_RUN.

PRODUCTION-INTENT - NOT RELEASED - NO HARDWARE RESULTS

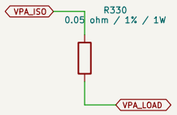
ARDF-TX

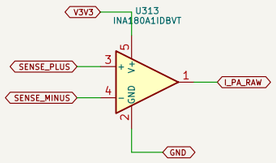
Sheet: /06b Independent threshold dividers/
File: 06b_thresholds.kicad_sch

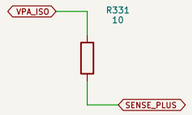
Title: 06b Independent threshold dividers

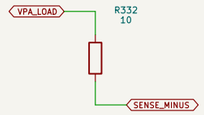
Size: A3	Date: 2026-10-18	**Rev: M4A-V1R1**
KiCad E.D.A. 10.0.6		Id: 8/15

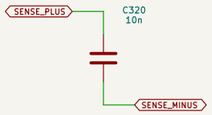
07 PA current trip and case-temperature interlock

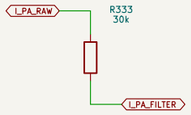


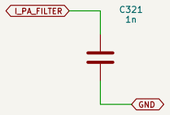


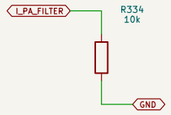


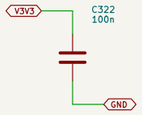


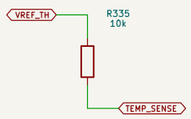


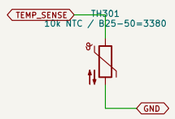


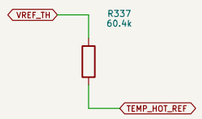


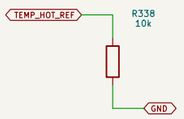


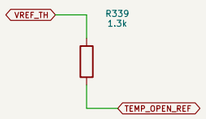


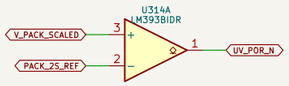


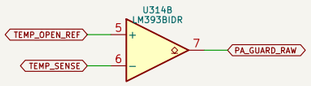


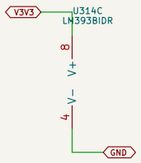


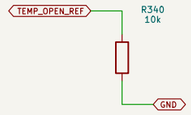


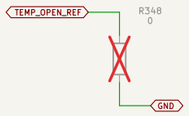


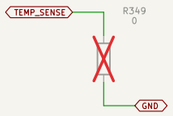


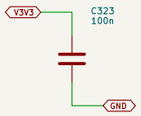














Current trip nominal1.251A (20*0.05ohm=1V/A); targetfaultmute<=100us; dividergain0.25V/A:7.5ustau.
Temperature window detects hot/short and detached/open sensor; nominalhot ~80C case.
Values/thermal contact, comparator tolerance, filtering and trip latency are qualification gates.

M4A-V1: PRODUCTION-INTENT - NOT RELEASED / all physical gates NOT_RUN.

PRODUCTION-INTENT - NOT RELEASED - NO HARDWARE RESULTS

ARDF-TX

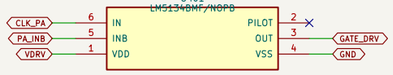
Sheet: /07 PA current trip and case-temperature interlock/
File: 07_current_thermal.kicad_sch

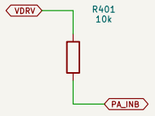
Title: 07 PA current trip and case-temperature interlock

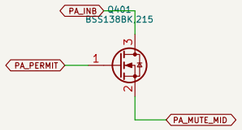
Size: A3	Date: 2026-10-18	**Rev: M4A-V1R1**
KiCad E.D.A. 10.0.6		Id: 9/15

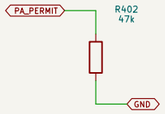
08 Proven normal RF core and QCX-derived drain shaping

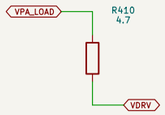
RF core adapted from OpenARDF SignalSlinger HW3.5 (MIT): shaper principle from QCX/W6JL.
Keep clock running during graceful key tail; INB immediately mutes on fault. This is NOT the VHF AM modulator.

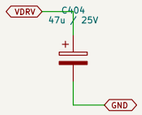


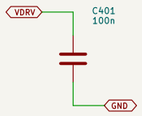


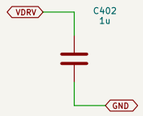


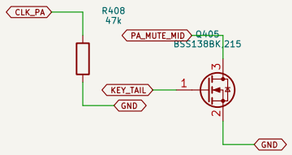


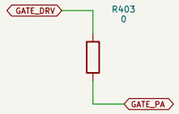


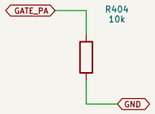


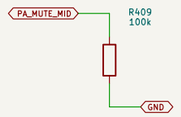






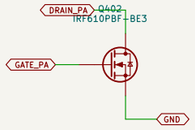


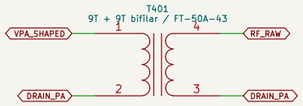


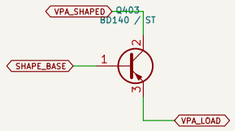


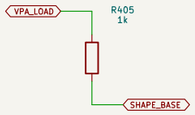


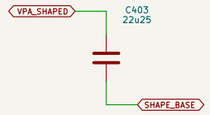
No PA/drain reservoir after shaper. Driver reservoir is upstream.
9+9 series autotransformer: effective voltage step2:1, load-12.5ohm beforefilter.
Transformer drawing is coupled coils; center join is external and explicit.
All components on this sheet DNP on Fox-only. No watt-to-Fox RF switching.

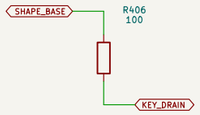


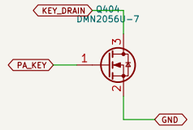


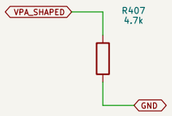














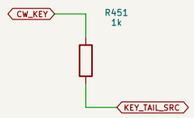
M4A-V1: PRODUCTION-INTENT - NOT RELEASED / all physical gates NOT_RUN.

PRODUCTION-INTENT - NOT RELEASED - NO HARDWARE RESULTS
ARDF-TX
Sheet: /08 Proven normal RF core and QCX-derived drain shaping/
File: 08_normal_rf.kicad_sch
Title: 08 Proven normal RF core and QCX-derived drain shaping

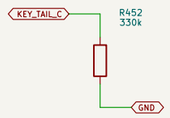
Size: A3	Date: 2026-10-18	**Rev: M4A-V1R1**
KiCad E.D.A. 10.0.6		Id: 10/15

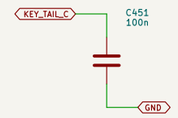
08b Shared key–tail and disarm reset

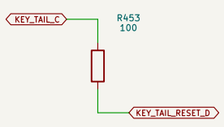
Path permission alone must not startRF during drain–integrator charging.
Normal series INB sink needsPA_PERMIT andKEY_TAIL; FoxRFmute also requiresFOX_PERMIT andKEY_TAIL.
Tail usesraw CW_KEY; disarm clampsmemorybelowSchmittlow. KEYhighcannotarm; capturedpath+withdrawaltrippreventsRFtransfer.

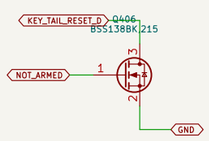












RCtailtau33ms; actualholddependsdiodedrop/HC14hysteresis. Benchqualify10–80mshold,
longerthanthe3–8msRFdecay. ShorterfallcanclipRF; excessiveholdraisesidle/leakage.
ARMlow>=20ms clearsmemory. Afterarming wait>=250msbeforefirstkey, thenverifysettleddrain.
A fault bypassesgracefuldecay: PApermission/driver/railoff immediately.
RFcalculation usesgatechargefortheentire20%scheduledslot, notonlyCWmarktime.

M4A-V1: PRODUCTION-INTENT – NOT RELEASED / all physical gates NOT_RUN.

PRODUCTION–INTENT – NOT RELEASED – NO HARDWARE RESULTS

ARDF–TX

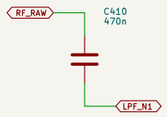
Sheet: /08b Shared key–tail and disarm reset/
File: 08b_key_tail.kicad_sch

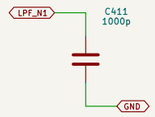
Title: 08b Shared key–tail and disarm reset

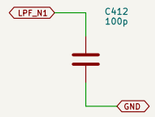
Size: A3	Date: 2026–10–18	**Rev: M4A–V1R1**
KiCad E.D.A. 10.0.6		Id: 11/15

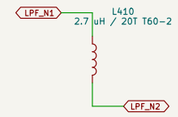
09 Original 80 m elliptic filter and normal port

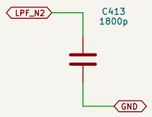
Retain SignalSlinger 80m values/topology: series470nF, two2.7uH, 1100/1800/1100pF,91pF acrosssecondcoil.
Split1100pF into1000+100pF for obtainable capacitors. Coil recipe is mandatory; no generic power inductors.

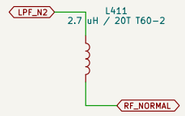


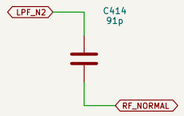


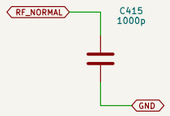


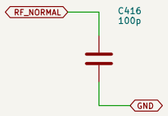


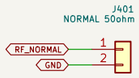














Reference50ohm power: J401, calibrated adapter+dummyload. No antenna gain/ERP conversion on80m.
Long antenna matcher remains separate: measured verticalwire+defined counterpoise, L-match/variometer.
Normal match START range15–50uH fromDF1FO: exacttap/cap values requireinstalledimpedance.
A screwterminal can save~4EUR versus Amphenol132134 SMA; reference-plane calibration remains mandatory.
Harmonics/spurs must be measured after this filter across3.510–3.600MHz and everycalibratedsetting.

M4A-V1: PRODUCTION-INTENT – NOT RELEASED / all physical gates NOT_RUN.

PRODUCTION-INTENT – NOT RELEASED – NO HARDWARE RESULTS

ARDF-TX

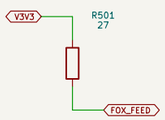
Sheet: /09 Original 80 m elliptic filter and normal port/
File: 09_normal_filter.kicad_sch

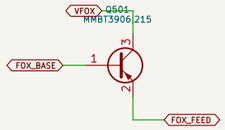
Title: 09 Original 80 m elliptic filter and normal port

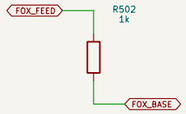
Size: A3	Date: 2026-10-18	**Rev: M4A-V1R1**
KiCad E.D.A. 10.0.6		Id: 12/15

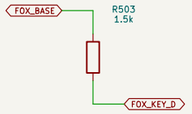
10 Independent Foxoring shaper and rail-following clock buffer

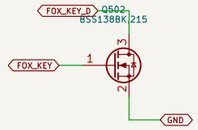
TinyFox three-gate mechanism with stronger SN74AC00DR; new rail-following NPN input + QCX-style low-current envelope.
No electrical RF connection to RF_RAW/RF_NORMAL. Nominal10mW is calibrated at separate J502, not inferred.

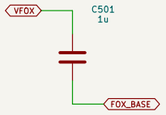


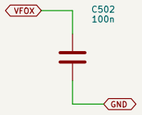


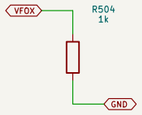


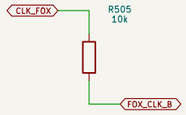


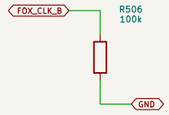


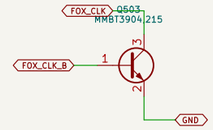


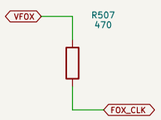




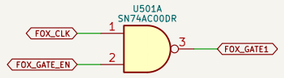


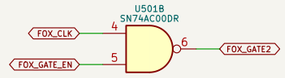


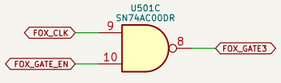


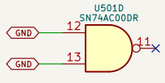


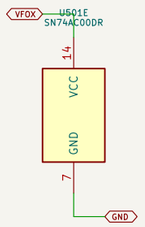






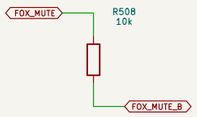


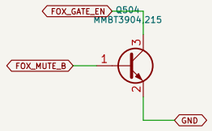


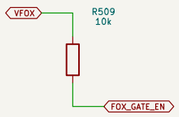












M4A-V1: PRODUCTION-INTENT - NOT RELEASED / all physical gates NOT_RUN.

PRODUCTION-INTENT - NOT RELEASED - NO HARDWARE RESULTS

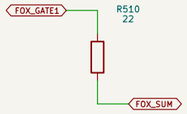
ARDF-TX

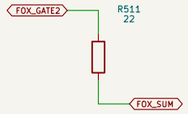
Sheet: /10 Independent Foxoring shaper and rail-following clock buffer/
File: 10_fox_rf.kicad_sch

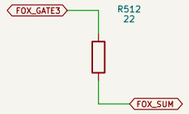
Title: 10 Independent Foxoring shaper and rail-following clock buffer

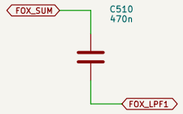
Size: A3	Date: 2026-10-18	**Rev: M4A-V1R1**
KiCad E.D.A. 10.0.6		Id: 13/15

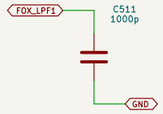
11 Separate Foxoring filter, calibration pad and protected compact port

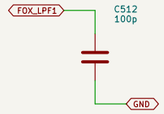


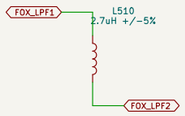


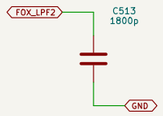


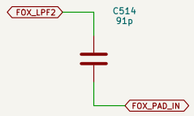


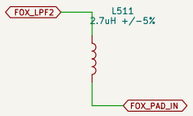


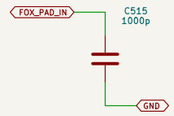


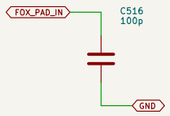


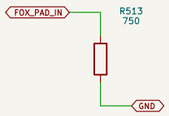


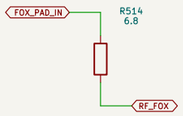


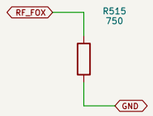


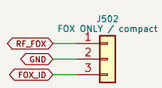


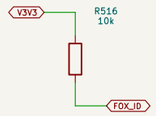


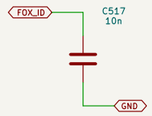














J502 reference plane: pin1 RF/pin2 definedreturn; use shorttwistedpair coaxadapter sharedfixture.
Compactaccessory: -30 cmradiator+definedcompactreturn; tune independentloading/couplingcartridge.
StartingTinyFoxreference100uH plus1.6–15pF: effectiveC5–15pF mayrequire134–402uH.
Exact30cm geometry/tap/coil/capvalues OPEN untilantenna impedance measurement; no Classicwire substitute.
Nominal10mW in50ohm and30maudible/250minaudible are TWO distinct acceptance tests.

M4A-V1: PRODUCTION-INTENT – NOT RELEASED / all physical gates NOT_RUN.

PRODUCTION-INTENT – NOT RELEASED – NO HARDWARE RESULTS

ARDF-TX

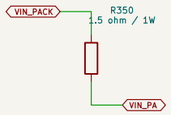
Sheet: /11 Separate Foxoring filter, calibration pad and protected compact port/
File: 11_fox_filter.kicad_sch

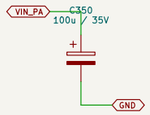
Title: 11 Separate Foxoring filter, calibration pad and protected compact por

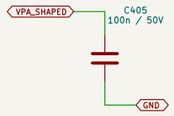
Size: A3	Date: 2026-10-18	**Rev: M4A-V1R1**
KiCad E.D.A. 10.0.6		Id: 14/15

12 Production test access input damper and low-capacitance probe access

M3B-E2: bounded12.0-16.8V protected4S only; first bring-up uses a CURRENT-LIMITED LAB SUPPLY.
R350/C350 damp cable L<=10uH; pack sense remains ahead of R350. TVS is NOT an18V clamp.
No driver fitted until power-only OV withdrawal test proves VDRV<=12.6V. No physical tests performed.



















Population stages are assembly instructions, not bypasses of hardware interlocks.
TH301 electrical sensor is remotely bonded to insulated PA case via test pads; do not fit a second NTC in parallel.
R403 zero-ohm can be removed to isolate the gate driver during dummy-load tests.
Do not inject28V into this board.16.8V failed-buck test uses rated surrogate load with U401/Q402 absent.

M4A-V1: PRODUCTION-INTENT - NOT RELEASED / all physical gates NOT_RUN.

PRODUCTION-INTENT - NOT RELEASED - NO HARDWARE RESULTS
ARDF-TX

Sheet: /12 Retained input damper and production test access/
File: 12_evaluation.kicad_sch

Title: 12 Retained input damper and production test access

Size: A3	Date: 2026-10-18	**Rev: M4A-V1R1**
KiCad E.D.A. 10.0.6		Id: 15/15